

Z.AI

YTL*AI LABS



PEKOM



UMHackathon 2026

SCADA-i (Supervisory Control and Data Acquisition Intelligence)

UMHackathon 2026 Submission (Deep-NaN)

Universities/colleges still rely on manual work when it comes to managing their facilities and amenities.



Fragmented management departments

Slow booking process



Students not switching off appliances

Huge energy waste



Reactive maintenance

Unexpected breakdowns



Tedious administrative work

Fail to meet compliances due to lack diligence

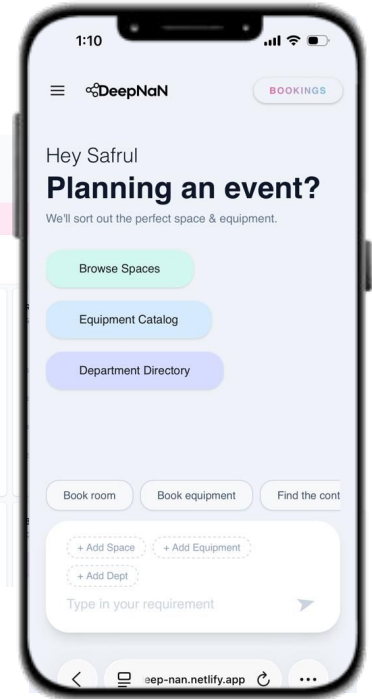
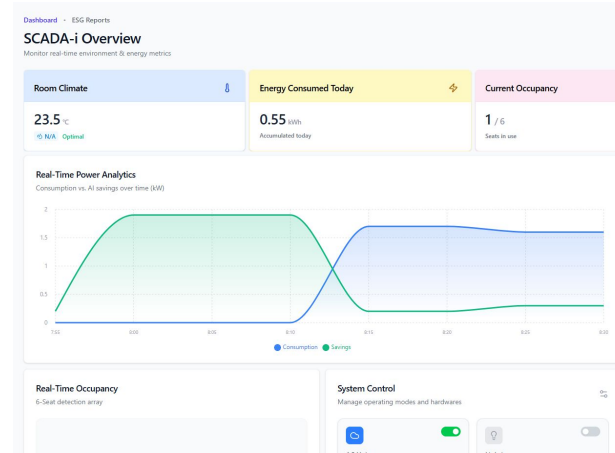
SCADA-i is your **university super managers!**

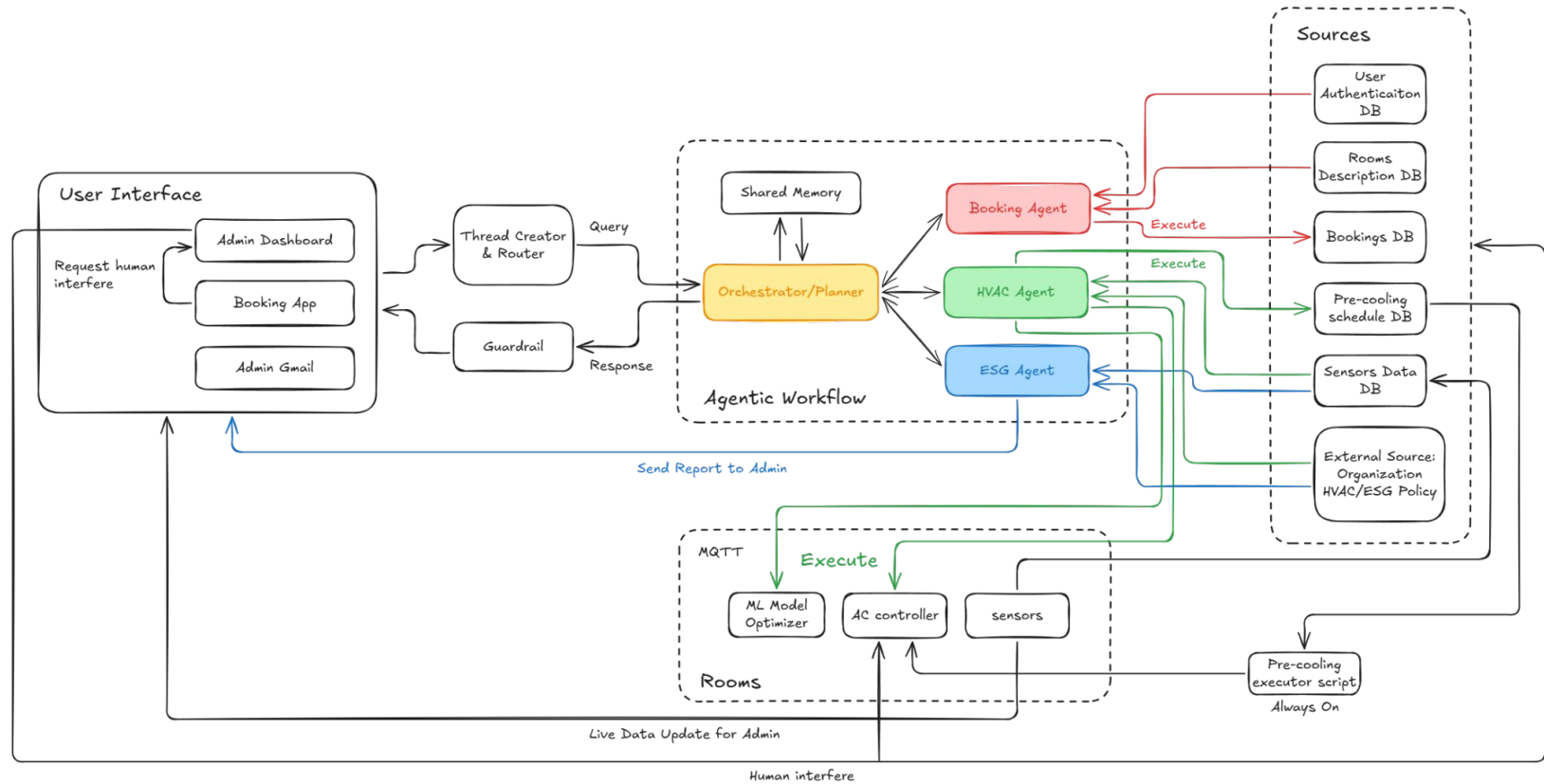
We manage the rooms,
We manage the bookings,
We manage the corporate reports.

What we offer:

- Multi-agent system
- Personalized mobile app
- Personalized dashboard

Better alignment within campus
— all within **one closed-loop system.**





Why Agents? No Lunch Breaks, No Blind Spots, No Wasted Energy.



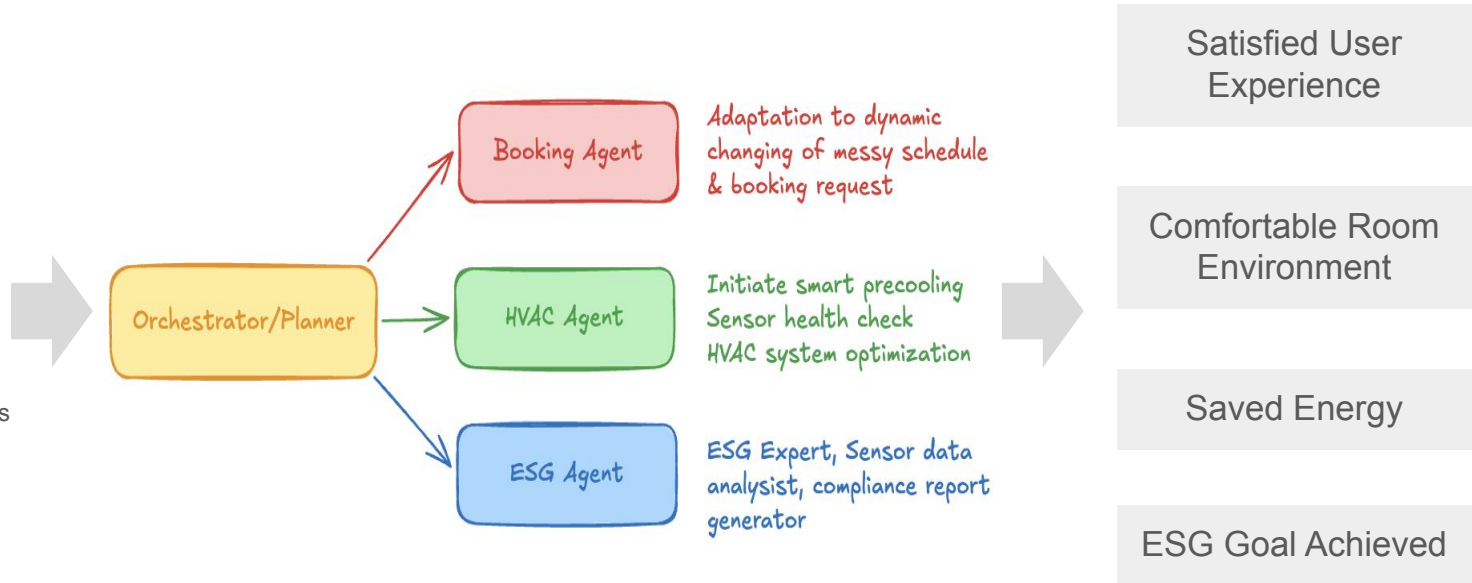
Messy Schedule, Booking Request



Empty classroom with air cond on / Unoptimized & Uncomfortable HVAC settings



Increasing concern of ESG

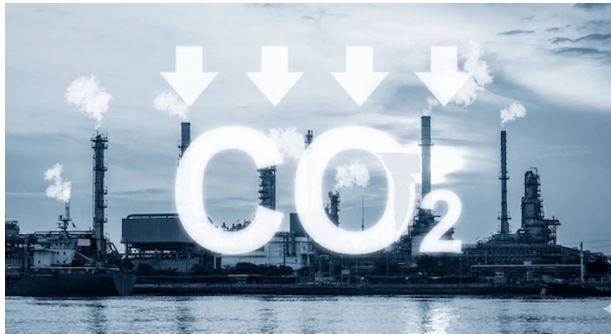


PER 4 CLASSROOM CLUSTER CALCULATION

Saves **RM6,100/yr**

13.44 MWh of energy reclaimed.

Covering initial capital investment in less than 3 months.

Avoids **10.40 tons of CO₂**

Directly aligns with UN SDGs and corporate ESG mandates.

Equivalent to planting ~475 trees every year per cluster.

Saves **40+ hrs/month**

Eliminates daily assessment, analysis and reporting.

Single orchestrator coordinates AI subagents end-to-end.

Onboarding & Integration

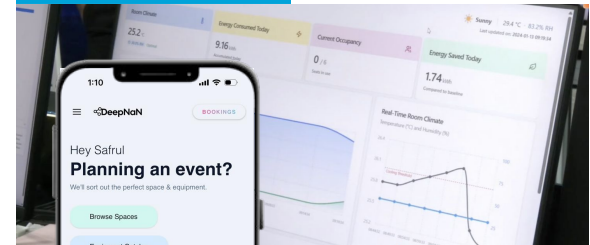
Complexity-based RM 15,000 - RM 50,000 / deployment for mapping existing IoT sensors and integrating with legacy academic timetables.

Hardware Partner Commission

10% - 15% commission of the order value when universities purchase necessary smart sensor upgrades through our referred local IoT vendors.

B2B SaaS Subscription & Managed Hosting

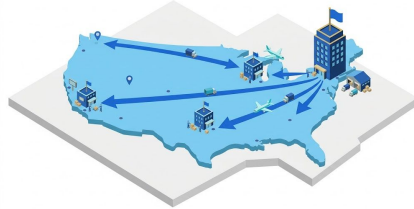
Approx. RM 2,500 - RM 25,000 / month per faculty or campus.
(Based on campus size and connected spatial zones)

The logo for Cytron Technologies, featuring the word "Cytron" in a stylized blue font with a swoosh underneath, and "Technologies" in a smaller blue font below it.The logo for DigiKey, featuring the word "DigiKey" in a bold, red, sans-serif font.

From Pilot to Global: Our 5-Year Growth Roadmap

Stage 1: Pilot Launch (Month 1-8)

Deploy in 3-5 key colleges/universities in Malaysia.



Stage 3: Regional Entry (Month 24-34)

Enter 3 neighboring countries with 1-3 universities/colleges selected per country.



Stage 2: National Expansion (Month 8-24)

Scale aggressively across MOHE networks (IPTA & IPTS).



Stage 4: SEA Dominance & Global Reach (Month 34-60)

Expand reach 50% of SEA market; prepare for global rollout



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That's all, Thank you for listening!

UMHackathon 2026 Submission (Deep-NaN)

1. Base Constants (Malaysian Industry Standards)

- **Electricity Tariff:** RM 0.454 / kWh (TNB Commercial Tariff B/C)
- **Grid Emission Factor:** 0.774 kg CO₂ / kWh (Energy Commission Peninsular Malaysia)
- **HVAC Power Density:** 80 W/m²
- **Total Power Density:** 80 W/m² (0.080 kW/m²)

2. Operational Assumptions (1 Single Meeting Room)

- **Room Size:** 35 m² (Standard university classroom / corporate boardroom)
- **Total Load per Room:** 35 m² × 0.080 kW/m² = 2.8 kW
- **Standard Operating Hours:** 12 hours/day × 250 working days/year = 3,000 hours/year
- **Baseline Annual Consumption:** 2.8 kW × 3,000 hours = 8,400 kWh/year (8.4 MWh)

3. The EcoPilot AI Savings (The 40% Rule)

- **Industry Context:** Research shows up to 40% of scheduled HVAC energy in meeting rooms is wasted due to ghost bookings, early dismissals, and running compressors at 100% for partial occupancy.
- **EcoPilot Intervention:** By utilizing YOLOv11n occupancy sweeps and LightGBM dynamic setpoints, the agent reclaims this 40% waste.
- **Energy Saved (1 Room):** 8,400 kWh × 40% = 3,360 kWh/year (3.36 MWh)
- **Financial Saved (1 Room):** 3,360 kWh × RM 0.454 = RM 1,525 / year
- **CO₂ Avoided (1 Room):** 3,360 kWh × 0.774 kg = 2,600 kg (2.60 Tons) / year

4. Scaling to a "4-Room Cluster" (Validating your Slide Numbers)

- **Total Energy Saved:** 3.36 MWh × 4 rooms = 13.44 MWh
- **Total Financial Savings:** RM 1,525 × 4 rooms = RM 6,100
- **Total CO₂ Avoided:** 2.60 Tons × 4 rooms = 10.40 Tons

Stage 1: MVP Rapid Prototyping

Deployed on lightweight, high-performance PaaS platforms (Netlify for React Frontend, Railway for Node/FastAPI Backend, and Supabase for PostgreSQL).



Stage 3: Enterprise Cloud Migration (Scaling Phase)

Migrating from fragmented PaaS to a unified, secure Virtual Private Cloud (VPC) on AWS or GCP.



Stage 2: System Hardening & Security

Strict Role-Based Access Control (RBAC) separating standard user access from administrative facility management, alongside user context injection to secure AI interactions.



Stage 4: Migrating from fragmented PaaS to a unified, secure Virtual Private Cloud (VPC) on AWS or GCP.

Hardware-as-a-Service (HaaS)

Multi-Tenant SaaS

Subscriptions (Standard to Premium) tracking AI and IoT metrics